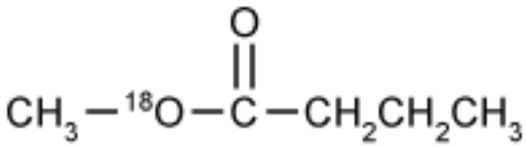
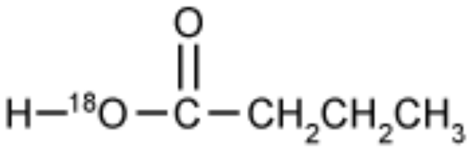
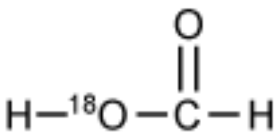
If the compound shown below is the product of a Fischer esterification, which of the following compounds is a starting reagent for the reaction and must contain oxygen-18?



A. $CH_3-{}^{18}OH$ [66%]

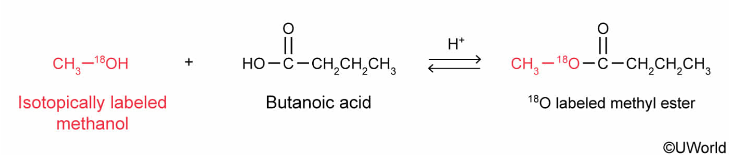
B. $H-{}^{18}O-CH_2CH_2CH_2CH_3$ [5%]

C.  [21%]

D.  [7%]

Correct
66% answered correctly

Explanation:



An atom in a compound can be substituted with an **isotope**, which can be tracked through a reaction. This procedure is called *labeling*. The final position of the isotopic label in the reaction product gives information about how the reaction occurs.

Esters are carboxylic acid derivatives that can be formed under acidic conditions through a Fischer esterification using a **carboxylic acid** and an **alcohol**. The reaction begins with protonation of the carbonyl oxygen of the carboxylic acid, followed by nucleophilic attack of the carbonyl carbon by the alcohol oxygen. A proton is then transferred from the alcohol oxygen to the hydroxyl group of the carboxylic acid, resulting in water (a good leaving group). Lastly, the carbonyl oxygen is deprotonated to yield an ester.

The ester given in the question is formed from butanoic acid and methanol. The isotopically labeled oxygen (^{18}O) in the ester is bonded to the carbonyl carbon and a methyl ($-CH_3$) group. Because the ester oxygen atom bonded to the carbonyl carbon and the alkyl group (methyl, in this instance) always **originates from** the **alcohol** reagent, the labeled starting material must be the alcohol. Therefore, the ^{18}O originates from ^{18}O-labeled methanol ($CH_3-^{18}OH$).

(Choice B) Although it is correct that the ^{18}O in the ester comes from an alcohol, this structure has too many carbon atoms. The four-carbon chain in the ester comes from a carboxylic acid (butanoic acid) rather than an alcohol.

(Choice C) It is true that the four-carbon chain originates from a carboxylic acid, but the ^{18}O in the ester must come from the alcohol rather than the carboxylic acid. If ^{18}O were in the hydroxyl group of the carboxylic acid prior to the reaction, then that ^{18}O atom would be found in a water molecule at the end of the reaction, not in the ester.

(Choice D) A carboxylic acid is involved in a Fischer esterification, but this structure has the incorrect number of carbon atoms, and the ^{18}O in the ester comes from the alcohol starting reagent.

Educational objective:
Esters are carboxylic acid derivatives that can be formed from a carboxylic acid and an alcohol through a Fischer esterification. The

ester oxygen bonded to both the carbonyl carbon and the alkyl group originates from the alcohol, a fact that can be confirmed by isotopically labeling the hydroxyl oxygen of the alcohol starting reagent.

Time spent:0 sec
Subject:
Organic Chemistry

QID:401605
System:
Functional Groups and Their Reactions